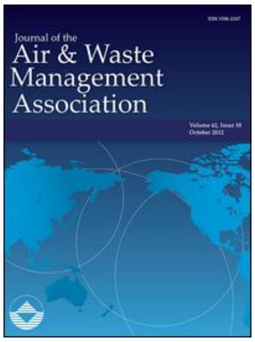

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Differences between Weekday and Weekend Air Pollutant Levels in Southern California

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ABSTRACT

Ambient air quality data were analyzed to empirically evaluate the effects of reductions of volatile organic compounds (VOCs) and oxides of nitrogen (NO_x) emissions on weekday and weekend levels of ozone (O_3; 1991–1998) and particulate NO_3^- (1980–1999) in southern California. Despite significantly lower O_3 precursor levels on weekends, 20 of 28 South Coast Air Basin (SoCAB) sites (28 of all 78 southern California sites) showed statistically significant higher mean O_3 levels on Sundays than on weekdays ($p < 0.01$); 49 of the remaining 50 sites showed no significant differences between mean weekday and Sunday peak O_3 levels. We also observed no statistically significant differences between mean weekday and weekend concentrations of particulate NO_3^- or nitric acid (HNO_3, the precursor of particulate NO_3^-). Averaged over sites, the mean Sunday NO_x and nonmethane hydrocarbon concentrations were 25–41% and 16–30% lower, respectively, than on weekdays. Site-to-site differences between weekend and weekday mean peak hourly O_3 levels were related to whether O_3 formation was limited by the availability of NO_x. A thermodynamic equilibrium model predicts that particulate NO_3^- levels would decrease in response to a reduction of HNO_3, and that particulate ammonium NO_3^- formation was not limited by the availability of ammonia. The similarity of mean weekday and weekend levels of NO_3^- therefore did not result from limitations on the formation of particulate NO_3^- from its precursor, HNO_3.

IMPLICATIONS

Reductions of NO_x and VOC emissions that are approximately equal to the difference between weekday and weekend rates of emissions may be ineffective or even counterproductive for reducing peak O_3 levels and particulate NO_3^- concentrations in southern California. Much larger (e.g., 80–90%) NO_x emission reductions than those that now occur on weekends (25–40%) could ultimately reduce O_3 concentrations, but compliance with ambient air quality standards may be delayed until such reductions have been fully implemented.

INTRODUCTION

In 1999, California's South Coast Air Basin (SoCAB) recorded 39 days with ambient ozone (O_3) concentrations exceeding the level of the federal 1-hr O_3 standard.[1] Particulate matter concentrations of less than 10-μm aerodynamic diameter (PM_{10}) exceeded the federal and state 24-hr PM_{10} standards on 6 and 258 days, respectively, during 1999. Particulate NO_3^-, which like O_3 is a secondary pollutant, is one of the principal contributors to both PM_{10} and fine particulate matter ($PM_{2.5}$) concentrations in the SoCAB, especially during the months of October through January.

Considerable effort has been expended to determine how responsive O_3 formation is to the control of emissions of volatile organic compounds (VOCs) or oxides of nitrogen (NO_x) in different urban and nonurban regions of the country. In many areas, photochemical air quality simulation models have been used to estimate the type and magnitude of emission controls needed to meet the requirements of the national ambient air quality standard for O_3. More recent model-development efforts include comprehensive gas-phase and particulate models. While such models are recognized as having the potential for quantitatively evaluating emission control strategies, model accuracies are unknown, and there have been notable examples where predictions have not been realized in the real world.[2–4] Therefore, increased attention has been devoted in recent years to the use of ambient aerometric data to empirically evaluate the response of secondary air pollutants to reductions of emissions of primary pollutants.

Because weekend O_3 and particulate NO_3^- precursor concentrations are significantly lower than weekday values at many locations, the comparison of weekend versus weekday primary and secondary species concentrations provides a means for empirically investigating the effects of VOC and NO_x reductions on O_3 and NO_3^- formation on a site-by-site basis.[5] The most important limitation of the empirical approach is that only the effects of actual emission changes can be investigated; the effects of much larger emission reductions than those that have occurred cannot be characterized.

The occurrence of typically higher peak O_3 values on weekends than on weekdays (the weekend effect) has been noted for many years and in a variety of locations.[6–10] It has been proposed that areas in which O_3 formation is VOC-limited tend to exhibit higher peak O_3 on weekends because of the more pronounced weekend reduction of NO_x than VOC emissions.[5] However, this interpretation of the data is not universally accepted. Other explanations for higher weekend peak O_3 values include possible higher precursor emissions on Friday evenings with carry-over to Saturday, increased emissions of VOC on weekends, timing of emissions on weekends, or increased reactivity of VOC emissions on weekends.[11,12] Careful examination of weekday and weekend precursor concentrations and composition is needed to correctly interpret the effects of weekend emissions on ambient O_3 and particulate NO_3^- levels.

In this study, ambient air quality data were analyzed to describe and evaluate weekday and weekend O_3 and particulate NO_3^- formation in southern California, focusing on the SoCAB. The data were used to characterize where or when weekend O_3 and particulate NO_3^- concentrations exceeded weekday levels and where or when weekend concentrations were lower than weekday levels. The statistical significance of the differences was evaluated using methods appropriate for the special statistical characteristics of the O_3 and NO_3^- concentration data. In keeping with common usage, "concentration" is used here in place of "mixing ratio" for gas-phase species. The units of measurement for O_3 and other gas-phase species discussed here are molar ratios in parts per billion by volume (ppbv) or parts per million by volume (ppmv). Units of measurement for particulate species are mass per unit volume (i.e., $\mu g/m^3$).

The differences between weekday and weekend ambient concentrations of O_3 precursor species (NO_x and nonmethane organic compounds [NMOCs]) and other primary pollutants (CO) were also characterized, and the statistical significance of these weekday–weekend differences was evaluated, again using statistical methods appropriate to the unique characteristics of the measurements.

Finally, the ambient measurements were used to evaluate where and when O_3 formation was limited by the availability of either NO_x or VOCs. For NO_3^-, additional analyses were carried out to characterize the relationships between particulate NO_3^- and its precursor species (NO_x, HNO_3, and NH_3).

METHODS

Statistical Methods

Tests for statistical significance of the weekend effect must be carefully chosen to be insensitive to serial correlation and seasonality in the pollutant data and to address the appropriate questions. Because O_3 standards focus on the extreme high values of the distribution, it is important to limit the test to high O_3 values. The focus on high O_3 days was achieved by selecting subsets of the data from each year's O_3 season (March 1–October 31). To determine the robustness of the results with respect to the choice of the data subset, 24 subsets were selected and compared. First, weekday averages were computed using either Monday–Friday or Tuesday–Thursday. Weekday averages computed using Tuesday–Thursday were indistinguishable from weekday averages computed using Monday–Friday. Some differences were observed among the means of the individual weekdays, however, with the Monday, Tuesday, and Wednesday means generally being lower than the Thursday or Friday means.

Second, for each site, the annual top 3, top 5, top 10, and top 2nd-through-11th daily peak hourly O_3 concentrations per day of week were selected for inclusion in the averages. The top 3 subset represents 21 days per year (~8.6% of all days), whereas the top 10 and top 2nd-through-11th subsets represent 70 days per year (~28.6% of all days). The mean weekday and weekend concentrations decreased as more days were included in each average. The top 3, top 5, top 10, and top 2nd-through-11th were highly correlated (r^2 of 0.98 or greater). There was no evidence that the means from any subset were substantially affected by large outliers. Averaged across sites, the top 5 means were about 90% of the magnitude of the top 3 means; the top 10 and top 2nd-through-11th means were about 75–80% of the magnitude of the top 3 means.

The statistical test for significant differences between average weekday and weekend peak O_3 levels is a form of permutation test.[13] Emission levels are lower on Sundays than on Saturdays, and graphical comparisons indicated that weekday–Saturday differences were smaller than weekday–Sunday differences. Therefore, the nonparametric test was used to test the statistical significance of the differences between Sunday and weekday peak O_3 concentrations at each site. The test is robust against non-normality and the presence of correlation among O_3 values on adjacent days (serial correlation).

To test the significance of the weekend effect in O_3 precursors with hourly data, a summary statistic was computed for each day (e.g., the value of the precursor at the time of peak O_3). Because the question with precursors is the extent to which weekend emissions differ, it was reasonable to consider all O_3 season days. For each week with complete data, the difference between the weekend mean and weekday mean was found. When there were at least 15 valid weeks, a sign test[14] was used. For those data measured once every 6 or every 12 days, including PM measurements, it was assumed that the values were statistically independent. Standard two-sample tests, namely, the *t* test and the Mann-Whitney test,[14] were applied.

DATA

Ozone and Precursor Data

The geographic domain studied was southern California. For the O_3 analyses, data were obtained for compliance monitoring sites located within five air basins, as defined by the California Air Resources Board (CARB): the South Coast, San Diego, South Central Coast, Salton Sea, and Mojave Desert air basins.[1] Ozone data were selected from the time periods 1991–1994, 1996–1998, and 1991–1998. These time periods are of interest because California Phase 2 reformulated gasoline was introduced in 1996, while federal Phase I reformulated gasoline was introduced at the beginning of 1995. To retain a focus on contemporary conditions, data before 1991 were not included. The O_3 weekend effect in southern California expanded from the 1980s to the 1990s.[12] Data from 1999 and 2000 were not yet available when these O_3 analyses were conducted.

PM and Precursor Data

For the NO_3^- analyses, both data from special studies and routine CARB data from the South Coast, South Central Coast, and Mojave Desert air basins for 1980–1999 were used. The entire period of record, 1980–1999, was used to enhance the statistical power of the comparisons because PM samples are collected with limited frequency and temporal resolution: PM_{10} and total suspended particulates (TSPs) are sampled once every 6 days, for a 24-hr period. The CARB particulate matter measurements include PM_{10} mass, PM_{10} NO_3^-, PM_{10} SO_4^{2-}, TSP mass, TSP NO_3^-, and TSP SO_4^{2-}, although SO_4^{2-} and NO_3^- concentrations were not reported for all PM and TSP monitoring locations. Twenty-four-hour averages (computed from hourly measurements) of O_3, CO, and NO_x were matched by date to the available PM_{10} and TSP measurements. The data were split by cool/wet (October–March) and warm/dry (April–September) seasons. The number of sites with at least 10 yr of PM data and at least 21 days per season per year (70% sampling completeness) were 9 PM_{10} and 26 TSP sites, of which there were 15 CO, 17 NO_x, and 25 O_3 monitoring sites. Because the NO_3^- measurements were limited in frequency, the analyses were not restricted to samples with high PM or PM NO_3^- concentrations.

The composite PM database also included multi-hour (usually 3-hr) hydrocarbon measurements for Photochemical Assessment Monitoring Stations (PAMS) for 1994 through 1999 as well as earlier CARB data for downtown Los Angeles. $PM_{2.5}$ data and other measurements were available for shorter time periods from the Carbonaceous Species Methods Comparison Study (CSMCS)[15] in 1986[16] and as later reanalyzed,[17] the 1987 Southern California Air Quality Study (SCAQS),[18,19] the California Acid Deposition Monitoring Program (CADMP) 1988–1994,[20] the 1995–1996 PM_{10} Technical Enhancement Program (PTEP),[21] and the 1997 South Coast Ozone Study (SCOS97).[22]

THE WEEKEND EFFECT FOR OZONE

Weekday and Weekend Levels of Ozone Precursor Species Concentrations

Ambient concentrations of NO_x, nonmethane hydrocarbons (NMHCs), NMOCs, selected NMOC species, and CO were obtained and analyzed. The primary purpose of this analysis was to establish that daytime ambient O_3 precursor concentrations were lower on weekends than on weekdays. (Although CO is not an O_3 precursor, it originates primarily from motor vehicle exhaust emissions and thus is a potentially useful indicator of motor vehicle exhaust VOC emissions.) Summary statistics were compiled for each monitoring location.

The statistics show that ambient VOC and NO_x concentrations were lower on weekends than on weekdays at the majority of sites and times of day (Table 1). For instance, depending upon the time of day, 74–76 of the 76 sites showed significantly lower NO_x concentrations on Sunday than on weekdays (see Table 1). Averaged over sites, the mean Sunday NO_x was 28% lower than the mean weekday NO_x concentration at the time of the peak O_3 and 25–41% lower at other times (Table 1 and Figure 1). Given the large number of NO_x sites and the consistency among sites, the estimated mean NO_x reductions are well quantified (see Figure 1).

The majority of sites with NMHC data showed significantly lower Sunday values than weekday values during one or more time periods (see Table 1). Averaged over sites, the mean Sunday NMHC was just 3% lower than the mean weekday NMHC at the time of peak O_3, but 16–30% lower at other times. On Saturdays, mean ambient NMHC concentrations were lower on weekends than on weekdays by approximately 1% at the peak O_3 hour and 10–22% at other times (see Table 1).

Ambient total NMOC and NMOC species concentrations were lower on weekends than on weekdays by 4–31% on Saturdays and 4–35% on Sundays (see

Table 1. Summary of comparisons between weekend and weekday mean primary pollutant concentrations during the O_3 season (March 1–Oct 31), 1991–1998, for all sites in five air basins. For split cell entries (|), two comparisons are shown: (1) the number of sites having lower (or higher) weekday means (left of |) and (2) the number of statistically significant ($p < 0.01$) results (right of |).[a,b]

Species	Sampling Period (start time to end time—PST)	Number of Sites: Sun. Mean < Weekday Mean	Sun. Mean > Weekday Mean	Sat. Mean < Weekday Mean	Sat. Mean > Weekday Mean	Percent Difference: Sunday Minus Weekday	Saturday Minus Weekday
NO_x (76 sites)	Peak O_3 hour	72|76[c]	4|0	73|76[c]	3|0	−28	−14
	6:00–9:00 a.m.	75|74	1|0	73|67	3|0	−41	−23
	9:00 a.m.–noon	74|76[c]	2|0	75|76[c]	1|0	−30	−14
	Noon–3:00 p.m.	75|76[c]	1|0	73|76[c]	3|0	−26	−16
	3:00–6:00 p.m.	73|75[c]	3|1	75|76[c]	1|0	−25	−20
CO (45 sites)	Peak O_3 hour	42|43[c]	5|4	30|29	17|18[c]	−12	+0.4
	6:00–9:00 a.m.	44|40	3|0	42|36	5|0	−32	−17
	9:00 a.m.–noon	44|45[c]	3|2	32|34[c]	15|13	−17	−3
	Noon–3:00 p.m.	42|43[c]	5|4	33|35[c]	14|12	−13	−4
	3:00–6:00 p.m.	45|45	2|2	42|42	5|5	−20	−13
NMHC (9 sites)	Peak O_3 hour	7|3	1|0	5|2	3|0	−3	−1
	6:00–9:00 a.m.	9|9	0	9|7	0	−30	−22
	9:00 a.m.–noon	9|8	0	8|6	1|0	−24	−15
	Noon–3:00 p.m.	8|6	1|0	7|4	2|0	−16	−10
	3:00–6:00 p.m.	9|7	0	8|6	1|0	−18	−11
NMOC[d] (2 sites)	5:00–8:00 a.m.	2|0	0	2|2	0	−30	−31
	Noon–3:00 p.m.	2[e]|2	0	2|0	0	−29	−10
Benzene (10 sites a.m., 8 sites p.m.)	5:00–8:00 a.m.	10|4	0	7|2	3|0	−31	−14
	Noon–3:00 p.m.	6|6	2|2	4|1	4|0	−25	−4
Toluene (10 sites a.m., 8 sites p.m.)	5:00–8:00 a.m.	10|6	0	7|3	3|0	−35	−17
	Noon–3:00 p.m.	8|8	0	6|1	2|0	−31	−9
Formaldehyde (5 sites a.m., 3 sites p.m.)	5:00–8:00 a.m.	5|0	0	4|1	1|0	−17	−16
	Noon–3:00 p.m.	1|1	2|2	3|0	0	−4	−12

[a]For NO_x, CO, and NMHC, the statistical test was the nonparametric Wilcoxon signed rank test.[14] This test was used in place of a standard *t* test because the *t* test is invalidated by the serial correlation of the hourly data. The nonparametric test ranks differences between each Saturday or Sunday and the average weekday value of the same week and is not invalidated by correlation between sequential days. For NO_x, CO, and NMHC, sites were included if they had at least 26 sample days for each day of the week (equivalent to 75% of one ozone season). Eight NO_x and two NMHC sites were excluded by this requirement; [b]For species other than NO_x, CO, and NMHC, the test was a two-sample *t* test. Samples were collected every third day so correlation between sequential days did not invalidate the test. Paired comparisons of Saturday or Sunday to same-week weekdays were not possible; [c]The nonparametric statistical test is not based on comparison of the means. It is possible for the number of sites with statistically significant lower weekend values to exceed the number of sites with lower weekend means because the nonparametric test was unaffected by outliers, whereas the means were sometimes influenced by a few high values. The inconsistency occurred at sites where a large majority of Saturday or Sunday values were lower than their corresponding weekday values, but the overall Saturday or Sunday mean was higher than the weekday mean because of the influence of a few weekend days having high concentrations; [d]PAMS reported total NMOC concentration. The data files for other PAMS sites did not report this value; [e]Excludes one high outlier on one Sunday at one site.

Table 1). However, the number of sites was limited and considerable variability occurred, so that the magnitudes of these reductions are not well quantified (see Figure 1).

CO levels averaged approximately 0–17% lower on Saturdays and approximately 12–32% lower on Sundays compared with weekdays. Statistically significant ($p < 0.01$) lower CO levels were observed at 42 of 47 sites on Saturdays and 45 of 47 sites on Sundays (see Table 1). Given the number of CO monitoring sites and the consistency and statistical significance of the weekend–weekday differences in ambient CO levels, these results support the more limited NMHC and NMOC data in indicating that daytime weekend ambient VOC levels were lower than corresponding weekday levels. Overall, daytime weekend VOC levels were lower than weekday levels, but the reduction was less than that which occurred for NO_x (see Figure 1).

Consistency of Ambient and Emissions Weekday–Weekend Precursor Differences

Weekend emissions inventories were unavailable from either the CARB or the South Coast Air Quality Management District, although a CARB inventory is in preparation.

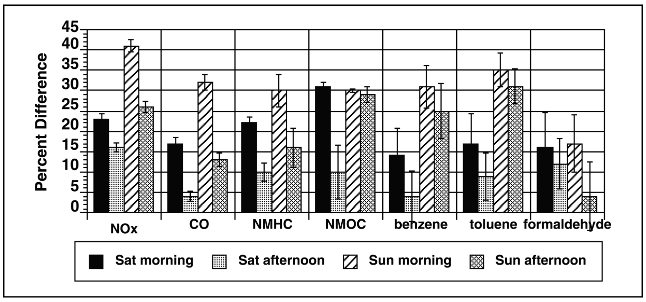

Figure 1. Average percentage differences in ambient concentrations of primary pollutants during four weekend time periods compared with corresponding weekday periods. Percentage differences were computed for each site and then averaged over sites. Error bars denote the standard error of the mean.

Companion papers[23,24] describe recent efforts to quantify weekend emissions rates. Survey and vehicle-count data both provide means for cross-checking the reasonableness of the observed differences between weekend and weekday ambient precursor concentrations. Survey results indicate that weekends exhibit a 12–18% reduction in VOC emissions and a 35–41% reduction in NO_x emissions relative to weekdays.[24] Much of the difference between weekdays and weekends is attributable to differences in motor vehicle–use patterns. In 2000, mobile sources accounted for 681 t/day of VOC emissions, while stationary and area sources contributed 210 and 200 t/day, respectively.[1] Also, mobile sources accounted for 1065 t/day of NO_x emissions, while stationary and area sources contributed 105 and 39 t/day, respectively.[1]

Vehicle-count data at 10 locations during the summer of 1997 show that heavy-duty truck (HDT) counts on Sundays ranged from ~20–60% of the Tuesday-through-Thursday counts.[24,25] Heavy-duty motor vehicles accounted for 27% of the total SoCAB NO_x emissions but less than 3% of VOC emissions in 2000.[11] Thus, the weekend reduction of HDT traffic potentially represents a reduction of ~10–20% of total NO_x emissions but less than ~1–2% of VOC emissions. Counts of non-HDT vehicles on Sundays were ~70–80% of the Tuesday-through-Thursday counts at the more central locations (Long Beach, Ventura Freeway, Van Nuys, Irvine, and Redlands) but were 95–180% at more distant sites (Fontana, Newhall, Castaic, Elsinore, and Indio).[25] Because non-HDT vehicles represented 39 and 53% of NO_x and VOC emissions, respectively, in 2000,[11] the reduced vehicle traffic in the more central portion of the basin would potentially provide an additional reduction of ~10% of NO_x emissions and ~13% of VOC emissions within the central SoCAB. The observed average ambient concentration reductions of ~25–40% for NO_x and ~10–30% for NMOC or NMHC are consistent with the vehicle-count and survey data.

Weekday and Weekend Ozone Levels

The average differences between Sunday and weekday peak O_3 concentrations were determined for the years 1991–1994, 1996–1998, or 1991–1998. The differences determined from these periods were highly correlated (r^2 of 0.83–0.88) and regression slopes were near unity. Therefore, only results for the full period, 1991–1998, are presented.

For the period 1991–1998, a nonparametric test was used to determine the statistical significance of the differences between Sunday and weekday peak O_3 concentrations at each site.[13] This nonparametric test is robust against non-normality and the presence of correlation among O_3 values on adjacent days (serial correlation). For each year at each site, the means of the daily averages of Sunday–Friday were ranked in ascending order. The possible ranks were 0–5. The ranks of the Sunday means were summed across years. For 8 years of data, the possible sum of Sunday ranks ranged from 0 to 40, attainable through a total of 6^8, or 1,679,616, possible combinations of ranks. A low-rank sum (<9 for 8 years of data) indicated that the Sunday mean was significantly ($p < 0.01$) lower than the mean of any weekday. A high-rank sum (>31 for 8 years of data) indicated that the Sunday mean was significantly ($p < 0.01$) higher than the mean of any weekday.

The statistical significance of the differences between Sunday and weekday peak O_3 concentrations showed some variation, depending on whether the differences were based upon the top 3, top 5, top 10, or top 2nd-through-11th peak O_3 values (per day of week per site per year). The top 3 data subset showed the fewest number of sites having significantly higher Sunday O_3; the top 2nd-through-11th data subset showed the largest number of sites having significantly higher Sunday O_3 (Table 2). Statistically lower mean Sunday peak O_3 occurred at one site in the top 3, top 5, and top 10 data subsets, and at no sites in the top 2nd-through-11th subset. The weekend effect (higher Sunday O_3) was most pronounced in the SoCAB (Figure 2 and Table 2).

Diurnal Profiles

It has been shown that levels of primary pollutants are lower on weekends than on weekdays, implying that O_3 precursor emission rates are lower on weekends than on weekdays. Because both VOC and NO_x emission rates are lower on weekends than on weekdays, the interpretation of the resulting changes in O_3 is more difficult than if only one precursor had been reduced. The effect of reduced concentrations of hydrocarbon species is to reduce the rate of radical formation, thereby reducing the rate of O_3 formation, whereas reduced concentrations of NO_x increase the rate of O_3 formation.[26] Reductions of

Table 2. Number of sites showing significantly higher or significantly lower ($p < 0.01$) Sunday mean O_3 compared with mean weekday O_3 (NS = not significant). The comparison was based on data from 1991–1998 and was carried out for each site having at least 6 yr of data. The data subsets were selected based on the number of high-O_3 days per day of week per year per site, as described in the text.

	Data Subset			
Air Basin	Top 3 Higher-NS-Lower	Top 5 Higher-NS-Lower	Top 10 Higher-NS-Lower	Top 2-11 Higher-NS-Lower
Mojave Desert	0-7-0	0-7-0	0-7-0	0-7-0
South Coast	14-13-1	17-11-0	19-9-0	20-8-0
South Central	0-30-0	1-29-0	2-28-0	4-26-0
San Diego	3-6-0	3-6-0	5-4-0	4-5-0
Salton Sea	0-4-0	0-3-1	0-3-1	0-4-0
Total	17-60-1	21-56-1	26-51-1	28-50-0

VOC emissions therefore either decrease peak O_3 concentrations, where O_3 formation is VOC-limited, or cause little change, where O_3 formation is NO_x-limited. In contrast, at locations where O_3 formation is strongly VOC-limited, reductions of NO_x emissions increase peak O_3 concentrations via reduction of O_3-quenching reactions (e.g., $NO + O_3 \rightarrow NO_2 + O_2$) and by enhancing radical-formation rates. Therefore, where NO_x emissions are significantly lower on weekends, weekend increases in mean peak O_3 are a possible indication of VOC limitation.[5] However, the combined reduction of VOC and NO_x concentrations may result in offsetting effects on peak O_3 concentrations at some sites. Moreover, differences in weekend and weekday O_3 concentrations might also result from other factors; thus, the data must be analyzed further to establish relations between precursor limitation and O_3 response.

Ozone cannot accumulate until essentially all the ambient NO has first been oxidized, because otherwise, NO reacts with O_3 to form NO_2 and O_2. The time at which O_3 can begin accumulating can be identified by plotting the difference, $O_3 - NO$, against time. The NO–O_3 crossover (the time when O_3 concentration equals NO concentration) is identifiable as the point at which the difference $O_3 - NO$ crosses zero.[12,27] Before this time, little O_3 is present, and the difference $O_3 - NO$ primarily reflects the ambient NO concentrations. After the crossover, little NO is present, and the difference $O_3 - NO$ primarily reflects O_3 concentrations. As shown in Figure 3, the NO–O_3 crossover occurred approximately 1 hr earlier on weekends than on weekdays at Los Angeles North Main and Azusa, sites typical of the central portions of the SoCAB. The morning emissions peak was evident on weekdays as the pronounced trough in the difference $O_3 - NO$ between 5:00 a.m. and 8:00 a.m. Conversion of the greater magnitudes of NO emissions on weekdays delayed O_3 production compared with weekends both on the highest O_3 days (see Figure 3) and on more moderate O_3 days.

At sites in the eastern portions of the SoCAB and in the Mojave Desert, peak O_3 concentrations were not significantly different on weekends than on weekdays, though the weekend precursor levels tended to be lower. Diurnal profiles of the difference $O_3 - NO$ reveal several characteristics of O_3 production at such sites (see Figure 3). At Santa Clarita (the only site with a statistically lower mean weekend peak O_3 for the period 1991–1998), the O_3–NO crossover was earlier on weekends than on weekdays, but the rate of O_3 formation was reduced enough on weekends compared with weekdays that the peak O_3 concentration was greater on weekdays (see Figure 3). Riverside showed diurnal profiles that were similar on weekends and weekdays, and the peak O_3 levels were not statistically different.

Comparison of Weekend Effects with Extent of Reaction

Ambient aerometric data have been used for qualitatively predicting whether O_3 formation at specific times and locations is VOC- or NO_x-limited.[28–34] Here, the extent of reaction computed using the Smog Production (SP) algorithm[13,34–36] is used to characterize VOC and NO_x limitation in southern California. In the CARB database, measurements of O_3, NO, and NO_x before 1994 were rounded to the nearest 10 ppbv. To avoid errors introduced by rounding, the calculations of extent of reaction and of changes in precursor concentration levels were limited to the period 1994–1998. The reporting of measurements to the nearest ppbv beginning in 1994 does not imply that the precision of the measurements improved, though in some cases, older instruments may have been replaced by newer ones.

Measurements of true NO_x ($NO + NO_2$) or NO_y (the sum of NO_x and NO_x reaction products) are not routinely available from compliance monitors. The instruments that are used in nearly all compliance monitoring networks measure NO_x plus unquantified fractions of NO_x reaction products, such as peroxyacetyl nitrate (PAN) and HNO_3.[37] We used these routine "NO_x" data to provide upper and lower bounds on the extent of reaction.[13] Because neither true NO_x nor true NO_y was measured, the bounds could not be compared with the true extent of

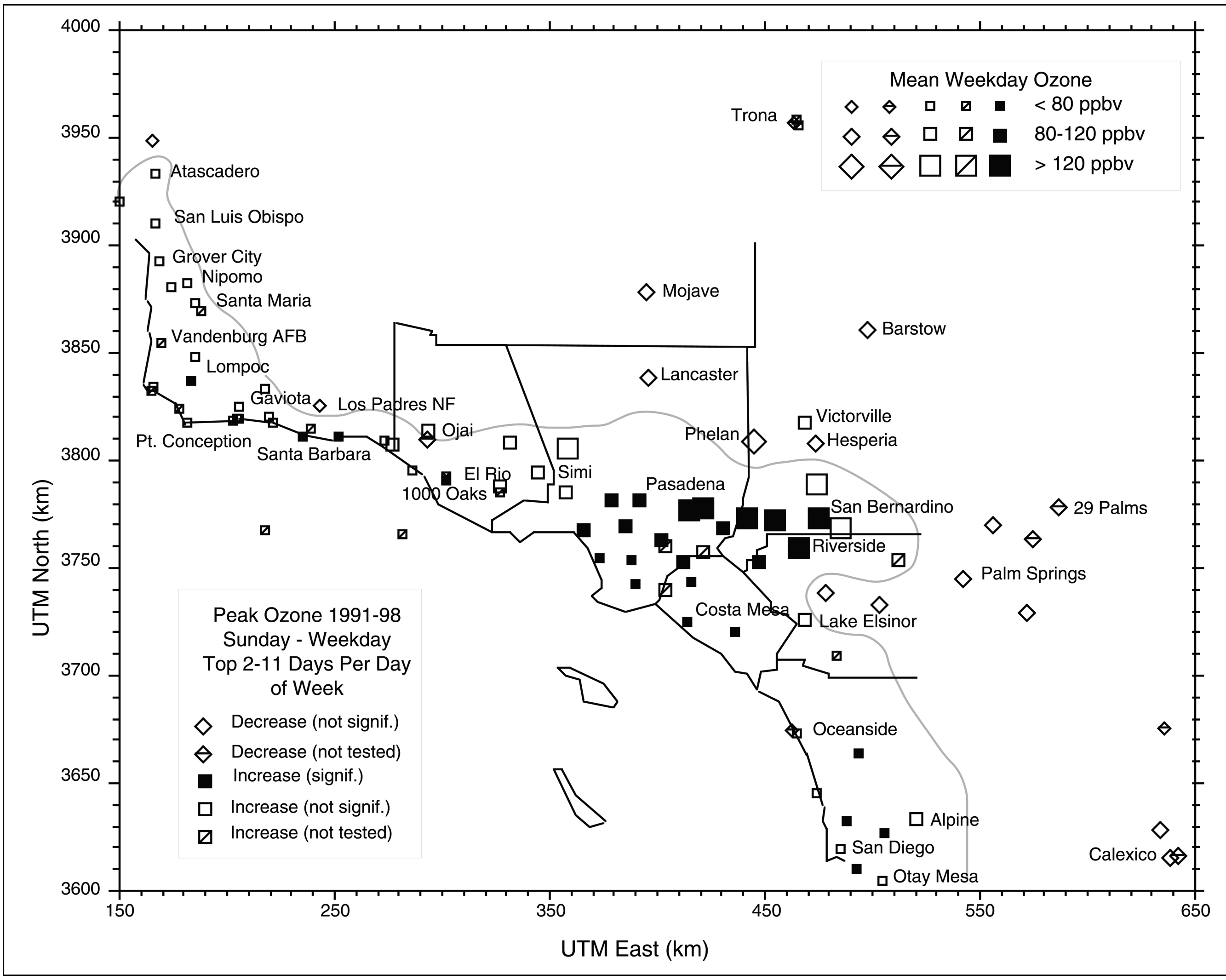

Figure 2. Geographic comparison of mean Sunday with mean weekday peak O_3 based on the top 2nd-through-11th days for each day of the week during each year, 1991–1998. Sites marked as increasing had higher mean Sunday peak O_3. The irregular line delineates an approximate transition between higher mean Sunday O_3 and higher mean weekday O_3.

reaction. Instead, the mean of the two bounds is used here as our best estimate of the true extent of reaction. For extents less than ~0.6, the bounds are very tight and the mean of the two bounds yields a good estimate of the true extent.[13] As extent increases, the bounds diverge. The lower bound is constrained to remain below 1, but the upper bound is not; it increases unrealistically to values exceeding 1. The uncertainties are greatest for mean extent of 0.6–0.9, which corresponds to conditions that are transitional between VOC and NO_x limitation.[36] However, when the mean extent is approximately 0.9, the bounds are ~0.7 and 1.1, so that mean extent above 0.9 indicates that the true extent of reaction is high and O_3 concentrations are usually NO_x-responsive.[36]

For each site, the O_3 weekend effect was compared with the mean extent of reaction (Figure 4). The range of peak O_3 concentrations among sites was large, so the results were expressed as fractional differences. The weekend effect was in part related to VOC or NO_x limitation of O_3 formation, and the results were reasonably consistent among the different data sets. The standard errors of the differences between mean weekend and weekday O_3 levels ranged from ~2 to 10 ppbv, so some of the scatter in Figure 4 is caused by the uncertainties of the mean weekday–weekend differences. The sites located in the South Coast, Mojave Desert, and Salton Sea air basins show a strong relation between differences in Sunday and weekday O_3 levels and the mean extent of reaction. For all basins, nearly all VOC-limited sites (extent <0.6) exhibited higher Sunday than weekday peak O_3 concentrations, whereas NO_x-limited sites (extent >0.9) did not exhibit consistent or significant weekend differences. The sites with the greatest increases in mean O_3 levels on Sundays all had mean extent less than 0.6. Thus, the interpretation of the sites' weekend–weekday O_3 differences as an indication of VOC or NO_x limitation[5] was supported.

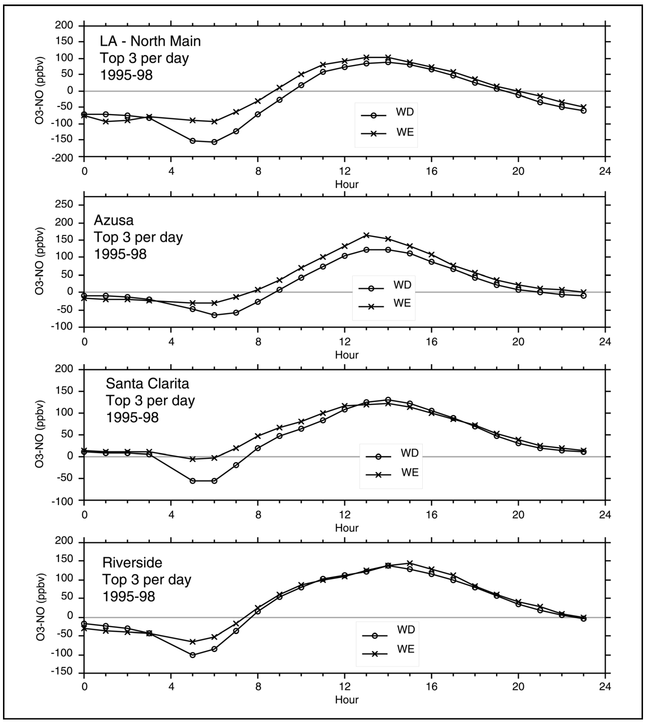

Figure 3. Weekday and weekend diurnal averages of the difference between O_3 and NO concentrations at North Main, Azusa, Santa Clarita, and Riverside. The averages were computed from the days having the three highest peak O_3 concentrations for each day of the week for the most recent years, 1995–1998.

THE WEEKEND EFFECT FOR PARTICULATE NITRATE

As is the case for O_3, particulate NO_3^- forms by means of a series of highly nonlinear atmospheric chemical reactions. Gas-phase oxidation of NO_2 yields HNO_3; aqueous-phase reactions are, by comparison, unimportant.[38] The rate of HNO_3 production is a nonlinear function of NO_x concentration, and may be limited by either hydroxyl radical (OH) or NO_2 concentrations or sunlight. Nitric acid reacts with NH_3 to produce particulate NO_3^-, which establishes a temperature- and humidity-dependent equilibrium with its gas-phase precursors. Sodium nitrate (and HCl) may also be generated via reaction of HNO_3 with NaCl in marine aerosol. The ambient concentrations of particulate NO_3^- are influenced by SO_4^{2-} concentrations, as particulate H_2SO_4 reacts preferentially with NH_3, relative to HNO_3. During seasons with lower temperatures and higher relative humidities, when equilibrium favors particulate NO_3^- over HNO_3 and NH_3, the concentrations of particulate NO_3^- may be limited by the availability of either NH_3 or HNO_3; specific situations must be studied to identify the limiting precursor.

Weekday and Weekend Differences in Ambient Species Concentrations

Because particulate NO_3^- derives from gas-phase precursors (HNO_3 and NH_3), data from the 1986 CSMCS, the SCAQS, the CADMP, and the 1997 Southern California Ozone Study (SCOS) were examined to characterize differences between weekday and weekend concentrations of HNO_3, PAN, NO_y, and O_3. These measurements were obtained over periods of 1–4 weeks' duration. Statistical tests of weekday–weekend differences were not applied, but graphical inspection of the data from these special studies indicates that mean concentrations of O_3 and PAN were higher on weekends than on weekdays (Figure 5). In contrast, mean weekend and weekday levels of HNO_3 did not necessarily exhibit differences: sometimes the hourly concentration profiles differed, but weekday and weekend concentrations were not different when averaged over 12- or 24-hr periods (see Figure 5). Thus, for measurements made at different times over a period of 12 yr, HNO_3 showed a weekend response different from either O_3 or PAN. However, mean weekend levels of HNO_3 were not lower than weekday concentrations, despite the observable reduction of NO_x concentrations on weekends compared with weekdays.

Formal statistical tests of differences between weekday and weekend levels of particulate NO_3^- and precursor species were carried out using data from the CADMP, the PTEP, and the routine compliance monitoring network. The CADMP operated three sites in southern California (Long Beach, downtown Los Angeles, and Azusa) from 1988 through 1994 on a once-per-six days, twice-per-day (6:00 a.m.–6:00 p.m. and 6:00 p.m.–6:00 a.m.) schedule. The CADMP sampler employed redundant measurements (both $PM_{2.5}$ and PM_{10} mass and speciation; two separate measures of particulate NO_3^- and of HNO_3). The measurements were compared to identify possible biases.[13] The

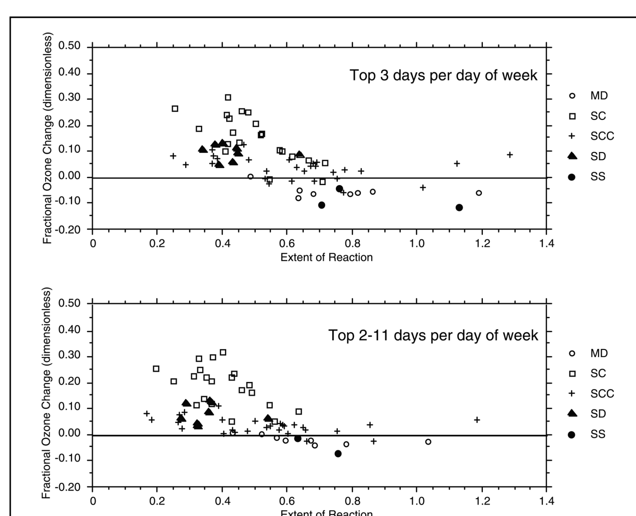

Figure 4. Comparison of 1991–1998 Sunday minus weekday peak hourly O_3 with the 1994–1998 mean weekday extent of reaction at the time of the O_3 peak, shown for top 3 and top 2nd-through-11th data subsets. Each symbol represents one monitoring location. Symbol types denote air basins.

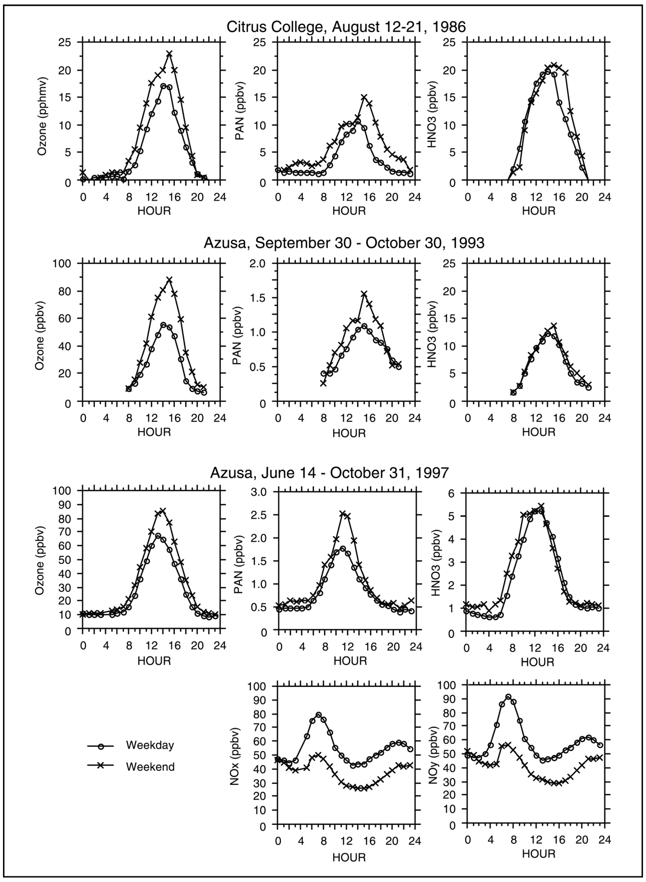

Figure 5. Weekday and weekend diurnal profiles of O_3, PAN, HNO_3, NO_x, and NO_y concentrations at Azusa during three time periods. During 1986, measurements were made at Citrus College, ~3 km east of Azusa.[16,17] During 1993, PAN was measured using the luminol method and HNO_3 was measured by tunable diode laser absorption spectroscopy (TDLAS).[17,46] During 1997, measurements of PAN were made June 14–October 16 by Daniel Grosjean Associates;[22] measurements of HNO_3 and of NO_y were made June 26–October 31 by CE-CERT.[22]

CADMP sampler was designed with a Teflon filter (for particulate NO_3^-) placed in front of a nylon filter (for HNO_3) in a sampling channel lacking an HNO_3 denuder; it had a separate channel with a nylon filter located downstream from an HNO_3 denuder. The comparisons indicated that the filter-pack HNO_3 (nylon filter, not denuded) was biased high by NH_4NO_3 volatilization from the Teflon filter. Correspondingly, the Teflon-filter $PM_{2.5}$ particulate NO_3^- measurements were biased low by loss. Volatilization was greatest during daytime hours in the warm/dry season. These comparisons indicate that routine PM NO_3^- measurements (which are made in a manner similar to that of the nondenuded CADMP Teflon filter) can be ~20–70% too low. Such losses introduce additional uncertainties into the comparison of weekend and weekday particulate NO_3^- levels.

All three CADMP sites show statistically significant ($p < 0.01$) daytime decreases of NO_2 from weekdays to weekends but no significant changes in HNO_3 (either as denuder difference HNO_3 or as HNO_3 without employing a denuder) or particulate NO_3^- (Table 3). At Azusa, statistically significant decreases in PM_{10} and $PM_{2.5}$ mass also occurred. The sites showed few significant differences at night. Therefore, the 1988–1994 CADMP data indicate that neither HNO_3 nor particulate NO_3^- levels were lower on weekends than on weekdays, even though precursor (NO_2) concentrations were significantly lower on weekends.

The PM measurements made during the 1995–1996 PTEP study show no statistically significant differences in NO_3^- concentrations at any of the five monitoring sites for either the $PM_{2.5}$ or PM_{10} size fractions (Figure 6). While the PM NO_3^- concentrations varied seasonally, no season shows significant differences between weekday and weekend mean NO_3^- levels.

The 1997 SCOS included four intensive sampling periods during which measurements were collected on both a weekday and a weekend day. These data were examined by pairing each Friday or Monday with the adjacent Saturday or Sunday. Because the number of sample points was so limited, the weekday and weekend values were compared without testing for statistical significance. At seven of the eight SCOS weekday–weekend pairs, morning NMOC concentrations decreased from weekday to weekend. NMOC/NO_x ratios increased on all four weekday–weekend pairs at Azusa, North Main Street, and Anaheim, and on two of three pairs at Burbank. Thus, these data indicate that differential decreases of VOC and NO_x emissions resulted in higher NMOC/NO_x ratios on weekends. Morning concentrations of NO_x, NO_y, NMOC, and CO were consistently lower on weekend days, while morning concentrations of O_3 and HNO_3 were higher on the majority of weekend days. Afternoon concentrations of O_3 were consistently higher on weekends, but the afternoon concentrations of HNO_3 were sometimes higher and sometimes lower on weekends.

A comparison of 24-hr mean weekday and weekend concentrations obtained from the routine compliance monitoring sites over their full periods of record shows that mean weekend concentrations were lower than weekday concentrations at statistically significant levels ($p < 0.01$) at 6 of 8 sites for PM_{10}, 12 of 15 sites for CO, 15 of 16 sites for NO_x, and 14 of 23 sites for TSP during the cool season (Figure 7). Weekend concentrations were higher than weekday concentrations at statistically significant levels for 14 of 23 sites for O_3 and 17 of 27 sites for the ratio of CO to NO_x. PM_{10} NO_3^- concentrations decreased at all sites from weekdays to weekends, though nearly all such decreases were not statistically significant (see Figure 7). Cool-season weekend TSP nitrate concentrations decreased at 12 sites and increased at 6 sites. The ratios of NO_3^--to-NO_x and NO_3^--to-PM_{10} mass were

Table 3. Comparison of mean daytime (6:00 a.m. through 6:00 p.m.) weekday with weekend concentrations at the three CADMP monitoring sites in the SoCAB. The significance level was determined from a two-sample *t* test, with values less than 0.01 (marked in bold) representing statistically significant differences.

Site	Season	Species	Mean Concentration (μg/m)		Significance Level
			Weekday	Weekend	
Los Angeles	Cool	HNO_3[a]	2.94 ± 1.44	3.24 ± 1.27	0.877
		HNO_3[b]	6.28 ± 0.51	7.75 ± 0.87	0.149
		NO_2	96.47 ± 4.78	71.09 ± 7.05	0.004
		PM_{10} NO_3	10.91 ± 1.03	10.19 ± 1.38	0.679
		$PM_{2.5}$ NO_3[a]	10.48 ± 1.47	16.69 ± 4.55	0.241
		$PM_{2.5}$ NO_3[b]	7.71 ± 0.83	7.42 ± 1.07	0.831
	Warm	HNO_3[a]	7.27 ± 1.52	10.59 ± 1.35	0.116
		HNO_3[b]	13.81 ± 0.69	14.91 ± 0.99	0.362
		NO_2	81.08 ± 5.17	58.54 ± 5.02	0.002
		PM_{10} NO_3	5.62 ± 0.34	5.51 ± 0.65	0.886
		$PM_{2.5}$ NO_3[a]	7.44 ± 1.06	6.00 ± 0.95	0.322
		$PM_{2.5}$ NO_3[b]	1.79 ± 0.21	1.57 ± 0.33	0.566
Azusa	Cool	HNO_3[a]	2.33 ± 0.67	5.31 ± 3.01	0.362
		HNO_3[b]	7.56 ± 0.56	8.51 ± 0.99	0.404
		NO_2	74.44 ± 5.13	52.94 ± 5.21	0.004
		PM_{10} NO_3	10.83 ± 1.15	7.76 ± 1.25	0.073
		$PM_{2.5}$ NO_3[a]	11.83 ± 1.68	11.37 ± 2.33	0.874
		$PM_{2.5}$ NO_3[b]	8.30 ± 0.93	5.25 ± 0.9	0.02
	Warm	HNO_3[a]	8.64 ± 1.65	10.78 ± 2.15	0.442
		HNO_3[b]	17.38 ± 0.81	15.72 ± 1.04	0.211
		NO_2	82.41 ± 4.7	44.85 ± 4.11	<0.001
		PM_{10} NO_3	5.62 ± 0.37	4.96 ± 0.56	0.328
		$PM_{2.5}$ NO_3[a]	7.77 ± 1.15	9.92 ± 1.61	0.294
		$PM_{2.5}$ NO_3[b]	1.82 ± 0.2	1.52 ± 0.33	0.439
Long Beach	Cool	HNO_3[a]	1.23 ± 0.3	1.21 ± 0.56	0.974
		HNO_3[b]	4.83 ± 0.37	5.41 ± 0.48	0.344
		NO_2	86.3 ± 5.34	80.75 ± 7.88	0.562
		PM_{10} NO_3	8.47 ± 0.87	9.52 ± 1.29	0.501
		$PM_{2.5}$ NO_3[a]	9.85 ± 0.86	10.36 ± 1.44	0.763
		$PM_{2.5}$ NO_3[b]	6.08 ± 0.69	6.71 ± 1.3	0.669
	Warm	HNO_3[a]	3.07 ± 0.2	3.54 ± 0.35	0.243
		HNO_3[b]	6.56 ± 0.4	6.63 ± 0.63	0.932
		NO_2	60.9 ± 3.05	41.88 ± 3.14	<0.001
		PM_{10} NO_3	3.92 ± 0.2	3.77 ± 0.33	0.689
		$PM_{2.5}$ NO_3[a]	4.22 ± 0.3	3.67 ± 0.39	0.261
		$PM_{2.5}$ NO_3[b]	0.81 ± 0.08	0.58 ± 0.05	0.021

[a]HNO_3 and $PM_{2.5}$ NO_3 from filters located downstream of an HNO_3 denuder; [b]HNO_3 and $PM_{2.5}$ NO_3 measurements were from filters with no HNO_3 denuders. Denuded samples marked as suspect[22] were excluded.

higher on weekends, although most differences were not statistically significant. At approximately half or more of the sites, the ratios of TSP NO_3^--to-NO_x and TSP NO_3^--to-TSP mass were significantly higher on weekends, which reflects the weekend decreases of NO_x and TSP mass at nearly all sites in combination with weekend increases of TSP NO_3^- at many sites. Comparisons for the warm season yielded nearly identical numbers of significant differences as for the cool season (see Figure 7).

In summary, neither routine nor special-study data reveal significant differences between weekday and weekend particulate NO_3^- or HNO_3 concentrations, despite significant decreases in ambient NO_x levels. These comparisons indicate a general lack of responsiveness of HNO_3 and particulate NO_3^- to changes in NO_x levels occurring from weekdays to weekends.

Equilibrium between Ammonia, Nitric Acid, and Particulate Nitrate

As previously noted, particulate NO_3^- formation is a nonlinear function of ambient NO_x levels. Conversion of NO_2 to HNO_3 is nonlinear, as is the equilibrium reaction of HNO_3 with NH_3 to form particulate NO_3^-. In this section, previous analyses of SCAQS and CADMP data,[39] which show that the availability of NH_3 generally does not limit the formation of particulate NO_3^- in the SoCAB, are confirmed. The method of analysis consists of application of a thermodynamic equilibrium model (SCAPE2)[40–44] in conjunction with examination of the ambient measurements.[39] Here, the measurements from the 1-yr PTEP study are analyzed.

SCAPE2 predicted the partition between particulate NH_4^+ and gas-phase N_3, and between particulate NO_3^- and gas-phase HNO_3, within about 5 μg/m^3 for most of the samples. In many cases, the agreement between predicted and measured values was even closer. The model tended to underpredict the particulate NH_4^+ concentrations and to overpredict the higher HNO_3 concentrations. Given the temporal resolution (24-hr) of the samples, and the temperature- and RH-dependent variations occurring between the gas and particle phases, better agreement between measurements and model predictions may be difficult to achieve. The samples that are thought to

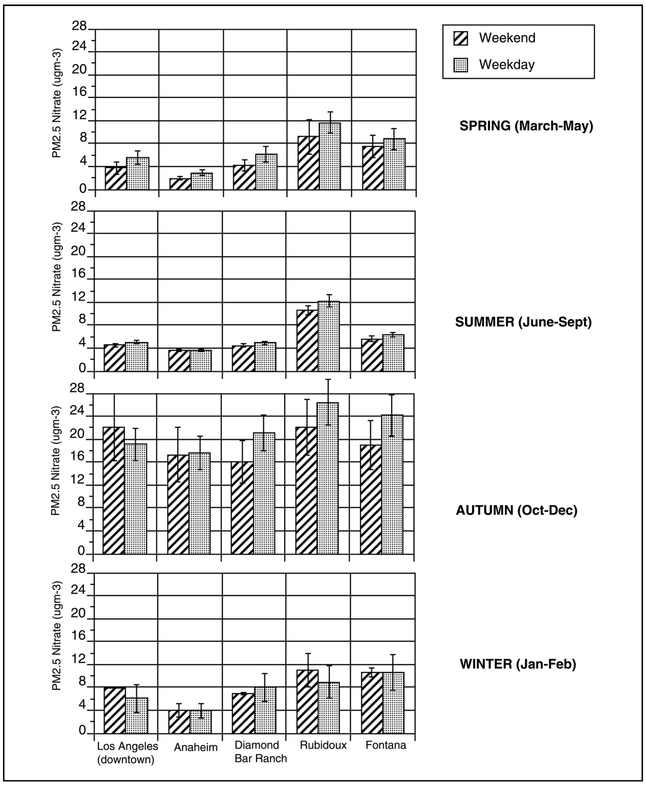

Figure 6. Mean seasonal weekday and weekend $PM_{2.5}$ nitrate concentrations at five locations during the 1995–1996 PTEP study. Error bars are one standard error of the mean.

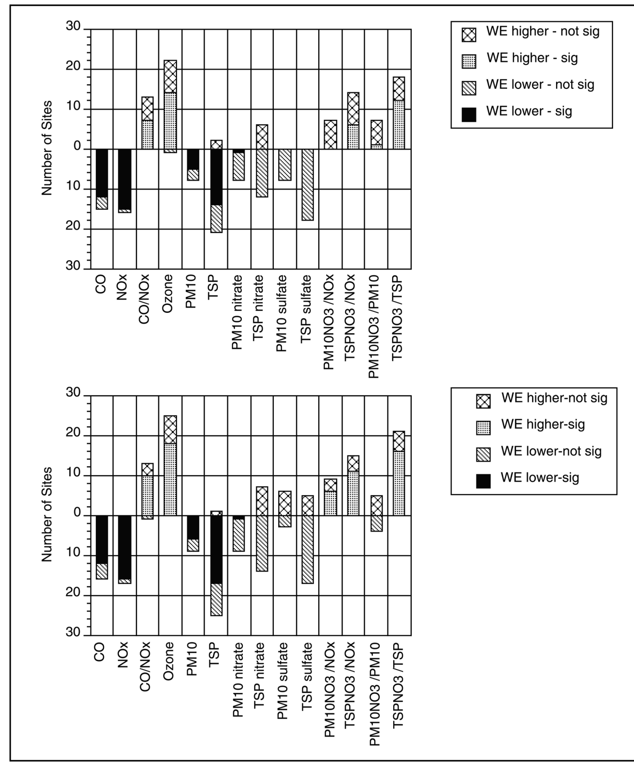

Figure 7. Number of routine compliance monitoring sites showing statistically significant differences ($p < 0.01$) in 24-hr mean weekday and weekend concentrations for the cool/wet season (October–March, top) and warm/dry season (April–September, bottom), 1985–1999. A t test was applied to the differences of the mean weekday and weekend concentrations for each year. The yearly tests were summed and approximated as a normal statistic.

be furthest from equilibrium were excluded from the analysis.[39,45]

Nearly all samples showed that particulate NO_3^- concentrations would decrease in response to a 20% reduction of HNO_3; in many cases, the predicted decrease was close to 20% (Figure 8). In contrast, predicted particulate NO_3^- concentrations decreased by much smaller amounts in response to a 20% decrease in NH_3 concentrations. Decreases in SO_4^{2-} concentrations left the particulate NO_3^- concentrations essentially unchanged. Thus, few samples showed any evidence of NH_3 limitation. Therefore, the similarity of mean weekday and weekend levels of particulate NO_3^- did not result from limitations on the formation of particulate NO_3^- from its precursor, HNO_3. Emission changes that lower the rates of formation of HNO_3 are therefore predicted to lower particulate nitrate concentrations as well.

CONCLUSIONS

Ambient environmental data were analyzed to describe and evaluate weekday and weekend O_3 and particulate NO_3^- formation in southern California. The comparison of weekend and weekday O_3 levels provided a means for empirically investigating the effects of VOC and NO_x reductions on O_3 formation, because the relative proportions and levels of O_3 precursor species were significantly different on weekends than on weekdays. Averaged over sites, the mean Sunday NO_x was 28% lower than the mean weekday NO_x concentration at the times of the peak O_3, and 25–41% lower at other times. Averaged over

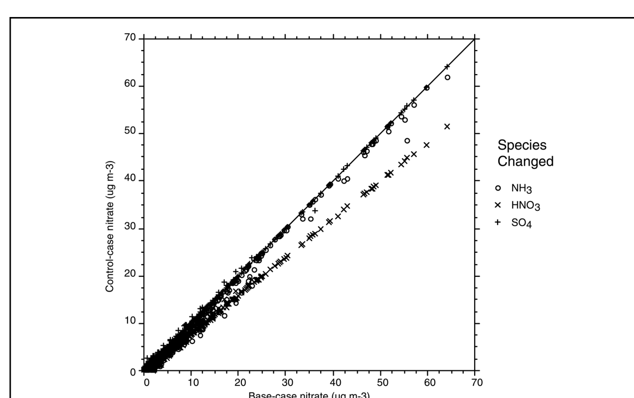

Figure 8. Emission reduction (control-case) vs. base-case NO_3^- concentration for 24-hr resolution data from the five sites of the PTEP study. The three emission-control cases (species concentration reductions) were 20% decreases of total NO_3^- (HNO_3 plus particulate NO_3^-), total NH_3 (NH_3 plus particulate NH_4^+), or SO_4^{2-}.

sites, the mean Sunday NMHC was 3% lower than the mean weekday NMHC at the times of peak O_3, and 16–30% lower at other times. Twenty of 28 SoCAB sites (28 of 78 total sites) showed statistically significant ($p < 0.01$) higher mean Sunday than weekday O_3 levels; 49 of the remaining 50 sites showed no significant differences between mean weekday and Sunday peak O_3 levels. Site-to-site differences between weekend and weekday mean peak hourly O_3 levels were related to their diurnal concentration profiles of O_3, NO, NO_x, and O_3-NO and whether O_3 formation was limited by the availability of NO_x. Nearly all VOC-limited sites exhibited higher weekend peak O_3 concentrations, whereas NO_x-limited sites did not show significant differences between mean weekday and weekend peak O_3 levels. Thus, the interpretation of the sites' responses as an indication of spatial patterns of VOC or NO_x limitation was supported.

Measurements from both routine monitoring and special studies during the years 1980 through 1999 showed no statistically significant differences between mean weekday and weekend particulate NO_3^- or HNO_3 (the precursor of particulate NO_3^-) concentrations, despite significantly lower weekend levels of NO_x. These comparisons indicate a general lack of responsiveness of HNO_3 and particulate NO_3^- to changes in NO_x levels occurring from weekdays to weekends. Application of a thermodynamic equilibrium model to data collected in 1995 and 1996 indicated that particulate NO_3^- levels would decrease in response to a reduction of HNO_3, and that particulate NH_4NO_3 formation was not limited by the availability of NH_3. The similarity of mean weekday and weekend levels of NO_3^- therefore did not result from limitations on the formation of particulate NO_3^- from its precursor, HNO_3. Emission changes that lower the rates of formation of HNO_3 are therefore predicted to lower particulate NO_3^- concentrations as well.

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REFERENCES

1. Alexis, A.; Delao, A.; Garcia, C.; Nystrom, M.; Rosenkranz, K. *The 2001 California Almanac of Emissions and Air Quality;* California Air Resources Board: Sacramento, CA, 2001.
2. Ingalls, M.N.; Smith, L.R.; Kirksey, R.E. *Measurement of On-Road Vehicle Emission Factors in the California South Coast Air Basin. Volume I. Regulated Emissions*; Final Report SCAQS-1, prepared for the Coordinating Research Council: Atlanta, GA, 1989.
3. Fujita, E.M.; Croes, B.; Bennett, C.L.; Lawson, D.R.; Lurmann, F.W.; Main, H.H. Comparison of Emission Inventory and Ambient Concentration Ratios of CO, NMOG, and NO_x in California's South Coast Air Basin; *J. Air & Waste Manage. Assoc.* **1992**, *42*, 264-276.
4. Lawson, D.R. "Passing the Test"—Human Behavior and California's Smog Check Program; *J. Air & Waste Manage. Assoc.* **1993**, *43*, 1567-1575.
5. Altshuler, S.L.; Arcado, T.D.; Lawson, D.R. Weekday vs. Weekend Ambient Ozone Concentrations: Discussion and Hypotheses with Focus on Northern California; *J. Air & Waste Manage. Assoc.* **1995**, *45*, 967-972.
6. Cleveland, W.S.; Graedel, T.E.; Kleiner, B.; Warner, J.L. Sunday and Workday Variations in Photochemical Air Pollutants in New Jersey and New York; *Science* **1974**, *186*, 1037-1038.
7. Lebron, F. A Comparison of Weekend-Weekday Ozone and Hydrocarbon Concentrations in the Baltimore-Washington Metropolitan Area; *Atmos. Environ.* **1975**, *9*, 861-863.
8. Graedel, T.E.; Farrow, L.A.; Weber, T.A. Photochemistry of the "Sunday Effect"; *Environ. Sci. Tech.* **1977**, *11*, 690-694.
9. Elkus, B.; Wilson, K.R. Photochemical Air Pollution: Weekend-Weekday Differences; *Atmos. Environ.* **1977**, *11*, 509-515.
10. Hoggan, M.; Hsu, M.; Kahn, M.; Call, T. *Weekday/Weekend Differences in Diurnal Variation in Carbon Monoxide, Nitrogen Dioxide, and Ozone—Implications for Control Strategies*. Presented at the 82nd Annual Conference & Exhibition of A&WMA, Anaheim, CA, June 1989; Paper 89-125.5.
11. *The Ozone Weekend Effect in California*; California Air Resources Board: Sacramento, CA, 2000. Available at: http://www.arb.ca.gov/aqd/weekendeffect (accessed May 2003).
12. Fujita, E.M.; Stockwell, W.R.; Campbell, D.E.; Keislar, R.E.; Lawson, D.R. Evolution of the Magnitude and Spatial Extent of the Weekend Ozone Effect in California's South Coast Air Basin, 1981–2000; *J. Air & Waste Manage. Assoc.* **2003**, *53,* 802-815.
13. Blanchard, C.L.; Fairley, D. Spatial Mapping of VOC and NO_x—Limitation of Ozone Formation in Central California; *Atmos. Environ.* **2001**, *35*, 3861-3873.
14. Lehmann, E. *Nonparametrics—Statistical Methods Based on Ranks;* Holden-Day: San Francisco, CA, 1975.
15. Lawson, D.R.; Hering, S.V. The Carbonaceous Species Methods Comparison Study—An Overview. *Aerosol Sci. Technol.* **1990**, *12*, 1-2; and entire issue of Vol. 12, no. 1.
16. Mackay, G.I.; Schiff, H.I. *Methods Comparison Measurements during the Carbonaceous Methods Comparison Study, Glendora, CA, August 1986: Tunable Diode Laser Absorption Spectrometer Measurements of HCHO, H_2O_2 and HNO_3*; Contract no. A5-189-32; California Air Resources Board: Sacramento, CA, 1987.
17. Tuazon, E.C.; Blanchard, C.L.; Hering, S.V.; Lucas, D.; Mackay, G.I. *A Review of Nitric Acid Measurements by Tunable Diode Laser Absorption Spectroscopy (TDLAS)*; Contract No. 93–340; California Air Resources Board: Sacramento, CA, 1995.
18. Lawson, D.R. The Southern California Air Quality Study; *J. Air & Waste Manage. Assoc.* **1990**, *40*, 156-165.
19. *Southern California Air Quality Study*; Proceedings of an International Specialty Conference, Los Angeles, California, July 1992; Air & Waste Management Association: Pittsburgh, PA, 1993.
20. Blanchard, C.L.; Michaels, H.; Tanenbaum, S. *Regional Estimates of Acid Deposition Fluxes in California for 1985–1994;* Contract No. 93-332; California Air Resources Board: Sacramento, CA, 1996.
21. Kim, B.M.; Lester, J.; Tisopulos. L.; Zeldin, M. Nitrate Artifacts during $PM_{2.5}$ Sampling in the South Coast Air Basin of California; *J. Air & Waste Manage. Assoc.* **1999**, *49*, PM142-PM153.
22. Fujita, E.M.; Green, M.; Keislar, R.; Koracin, D.; Moosmüller, H.; Watson, J. *SCOS97-NARSTO 1997 Southern California Ozone Study and Aerosol Study*; Contract No. 93-326; California Air Resources Board: Sacramento, CA, 1999.
23. Fujita, E.M.; Campbell, D.E.; Zielinska, B.; Sagebiel, J.C.; Bowen, J.L.; Goliff, W.S.; Stockwell, W.R.; Lawson, D.R. Diurnal and Weekday Variations in the Source Contributions of Ozone Precursors in California's South Coast Air Basin; *J. Air & Waste Manage. Assoc.* **2003**, *53,* 844-863.

24. Chinkin, L.R.; Coe, D.L.; Funk, T.H.; Hafner, H.R.; Roberts, P.T.; Ryan, P.A.; Lawson, D.R. Weekday versus Weekend Activity Patterns for Ozone Precursor Emissions in California's South Coast Air Basin; *J. Air & Waste Manage. Assoc.* **2003**, *53,* 829-843.
25. Larsen, L. *Day of Week Patterns of Heavy-Duty and Non-Heavy Duty Vehicle Activity at Weigh-in-Motion (WIM) Stations Relevant to the South Coast Air Basin During the Summer of 1997;* California Air Resources Board: Sacramento, CA, 1999. Available at: http://www.arb.ca.gov/aqd/weekendeffect (accessed May 2003).
26. Whitten, G.Z. The Chemistry of Smog Formation: A Review of Current Knowledge; *Environ. Int.* **1983**, *9*, 447-463.
27. Fujita, E.M.; Stockwell, W.R.; Keislar, R. *Weekend/Weekday Observations in the South Coast Air Basin;* Contract No. ACI-0-29086-01; National Renewable Energy Laboratory: Golden, CO, 2000. Available at: http://www.arb.ca.gov/aqd/weekendeffect (accessed May 2003).
28. Trainer, M.; Parrish, D.D.; Buhr, M.P.; Norton, R.B.; Fehsenfeld, F.D.; Anlauf, K.G.; Bottenheim, J.W.; Tang, Y.Z.; Wiebe, H.A.; Roberts, J.M.; et al. Correlation of Ozone with NO_y in Photochemically Aged Air; *J. Geophys. Res.* **1993**, *98*, 2917-2926.
29. Sillman, S. The Use of NO_y, H_2O_2, and HNO_3 as Indicators for Ozone-NO_x-Hydrocarbon Sensitivity in Urban Locations; *J. Geophys. Res.* **1995**, *100* (D7), 14175-14188.
30. Chang, T.Y.; Suzio, M.J. Assessing Ozone-Precursor Relationships Based on a Smog Production Model and Ambient Data; *J. Air & Waste Manage. Assoc.* **1995**, *45*, 20-28.
31. Cardelino, C.A.; Chameides, W.L. An Observation-Based Model for Analyzing Ozone Precursor Relationships in the Urban Atmosphere; *J. Air & Waste Manage. Assoc.* **1995**, *45*, 161-180.
32. Cardelino, C.A.; Chameides, W.L. The Application of Data from Photochemical Assessment Monitoring Stations to the Observation-Based Model; *Atmos. Environ.* **2000**, *34*, 2323-2332.
33. Chang, T.Y.; Chock, D.P.; Nance, B.I.; Winkler, S.L. A Photochemical Extent Parameter to Aid Ozone Air Quality Management; *Atmos. Environ.* **1997**, *31*, 2787-2794.
34. Blanchard, C.L.; Lurmann, F.W.; Roth, P.M.; Jeffries, H.E.; Korc, M. The Use of Ambient Data to Corroborate Analyses of Ozone Control Strategies; *Atmos. Environ.* **1999**, *33*, 369-381.
35. Blanchard, C.L. Ozone Process Insights from Field Experiments—Part III: Extent of Reaction and Ozone Formation; *Atmos. Environ.* **2000**, *34*, 2035-2043.
36. Blanchard, C.L.; Stoeckenius, T. Ozone Response to Precursor Controls: Comparison of Data Analysis Methods with the Predictions of Photochemical Air Quality Simulation Models; *Atmos. Environ.* **2001**, *35*, 1203-1215.
37. Winer, A.W.; Peters, J.W.; Smith, J.P.; Pitts, J.N., Jr. Response of Commercial Chemiluminescent NO-NO_2 Analyzers to Other Nitrogen-Containing Compounds; *Environ. Sci. Technol.* **1974**, *8*, 1118-1121.
38. Seinfeld, J.H.; Pandis, S.N. *Atmospheric Chemistry and Physics: From Air Pollution to Climate Change;* Wiley & Sons: New York, 1998.
39. Blanchard, C.L.; Roth, P.M.; Tanenbaum, S.J.; Ziman, S.D.; Seinfeld, J.H. The Use of Ambient Measurements to Identify which Precursor Species Limit Aerosol Nitrate Formation; *J. Air & Waste Manage. Assoc.* **2000**, *50*, 2073-2084.
40. Kim, Y.P.; Seinfeld, J.H.; Saxena, P. Atmospheric Gas-Aerosol Equilibrium I. Thermodynamic Model; *Aerosol Sci. Technol.* **1993**, *19*, 157-181.
41. Kim, Y.P.; Seinfeld, J.H.; Saxena, P. Atmospheric Gas-Aerosol Equilibrium II. Analysis of Common Approximations and Activity Coefficient Calculation Methods; *Aerosol Sci. Technol.* **1993**, *19*, 182-198.
42. Kim, Y.P.; Seinfeld, J.H. Atmospheric Gas-Aerosol Equilibrium III. Thermodynamics of Crustal Elements Ca^{2+}; K^+; Mg^{2+}; *Aerosol Sci. Technol.* **1995**, *22*, 93-110.
43. Meng, Z.; Seinfeld, J.H.; Saxena, P.; Kim, Y.P. Atmospheric Gas-Aerosol Equilibrium IV. Thermodynamics of Carbonates; *Aerosol Sci. Technol.* **1995**, *23*, 131-154.
44. Meng, Z.; Seinfeld, J.H.; Saxena, P.; Kim, Y.P. Gas/Aerosol Distribution of Formic and Acetic Acids; *Aerosol Sci. Technol.* **1995**, *23*, 561-578.
45. Wexler, A.S.; Seinfeld, J.H. Analysis of Aerosol Ammonium Nitrate: Departures from Equilibrium during SCAQS; *Atmos. Environ.* **1992**, *26A*, 579-591.
46. Mackay, G.I. *Measurement of Atmospheric Nitric Acid at Azusa by Tunable Diode Laser Absorption Spectrometry*; Contract No. 93-300; California Air Resources Board: Sacramento, CA, 1994.

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